

Data Storytelling

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Spring 2026

Week 10

Announcements

- Your proposal is due tonight! At 11:59 PM. Please submit it to me *via email*.
 - Register your group in the shared Google Sheet
- Homework 2 will be assigned next week
- Grading is coming along

Data Collection

Ethics and Legality of Data Collection

Data Collection

- Data is in many ways more abundant than ever.
 - What constraints do we face when gathering data?
 - What techniques can be used to gather data?
- But more actors are engaged in data collection than ever before.
 - Large language models require massive volumes of text data for training.
 - This increased traffic has posed a growing burden on websites

Data Sources

- **Public Data**

- Many academic and governmental institutions provide large volumes of data for free
- Ex. US FED, World Bank, V-Dem, public social media posts, aggregated search history

- **Subscription Services**

- There are many databases that are selectively made available to those who have institutional affiliations

- **Private Data**

- Data that was created with the expectation of privacy
- Person-to-person messages, internal corporate communications, individual search history

The Law and Data Collection

- *HIPAA*
 - Collecting or distributing Protected Health Information (PHI) is highly regulated and subject to strict punishment if misused.
 - Fines up to \$250,000 and 10 years in prison
 - Subjects *must* provide consent for PHI to be shared or used for research purposes

The Law and Data Collection

- *Fair Use*
 - A copyright doctrine that authorizes copyrighted works to be used without authorization under certain circumstances
 - So long as you are scraping public data and transforming it in some way, i.e. using it for research or analysis, it is exempted from copyright protections
 - *Does not apply if the goal is to sell the copyrighted data with no transformation for profit*

Ethical Considerations

- **Terms of Service (TOS)**

- Always review the terms of service for any platform or data source you intend to use. Violating these terms can lead to legal consequences and ethical issues.
- Failure to adhere to TOS can lead to bans from platforms

- **Privacy and Consent**

- Ensure that you have the necessary permissions to collect and use data, especially when it involves personally identifiable information (PID).

Practical Considerations

- Oftentimes, TOS are not clear, leading to grey areas
- Aggressive data collection, even when legal, can lead to your IP address getting blocked or having your access to subscription data suspended
- Social media platforms have recently adopted highly restrictive APIs
 - Response to high demand from AI companies

Example 1: Twitter/X

- X is still a popular platform among public figures, used to discuss current events
- API at one point charged \$200 per month to access ~2,000 posts per month
 - Current X Pricing

Example 2: ProQuest

- ProQuest is an online subscription database that Stevens students can access.
 - Searchable database of newspapers
- You may manually download news articles, ~20-100 at a time. Automating this process is a violation of its Terms of Use
 - Violation of TOU, can result in your account being suspended
 - Is possible to collect data using ProQuest API, which is recommended

Best Practices

- If collecting large volumes of data throttle your collection speed to avoid overloading the target's server
- Use APIs whenever possible
- If your data includes PID anonymize it
- If necessary, consult with your institutions' *Internal Review Board*, particularly if you need to gather data in a survey or experimental setting

Laws and Ethics involving other States

- China's Data Security Law
 - All data that touches a Chinese server requires approval from the Chinese government to be shared with foreign entities
- General Data Protection Regulation (GDPR)
 - Publicly available information is not automatically permitted to be scraped
 - Scraping sensitive information – health, political views, private life – is largely forbidden

Utilizing Existing Data

Public Datasets

- Data is often available for free from various sources, such as government agencies, research institutions, and non-profit organizations.
 - Examples include:
 - U.S. Census Bureau
 - World Bank
 - Kaggle Datasets
 - Google Dataset Search
- These datasets can provide valuable insights and save time compared to collecting data from scratch.

Things to keep in mind when using public data

- **Check source credibility**

- Data from reputable sources, known institutions with a track record of accuracy and reliability, is more likely to be trustworthy.
- Some sources may have low quality standards, biases, or agendas that can affect the integrity of the data.
- Good: U.S. Census Bureau, World Bank, Pew Research Center, etc.
- Bad: Official data from North Korea

- **Read the documentation**

- Illustrates how variables were coded and collected, and any limitations or caveats associated with the data.

Google Dataset Search

- A public searchable database of databases
- A great resource for exploring publicly available datasets

World Bank Data

- One of the best resources for exploring national-level economic data
- Contains everything from GDP and inflation to trade data
- *Notes:*
 - Does not contain data for all countries for all variables
 - Data is self-reported, meaning there is some assumed trust in national-level governments

V-Dem

- A great source of data on country-level political institutions
- Includes polyarchy, a measure of the degree to which a country is democratic or authoritarian
- Metrics are based on surveys of thousands of area and political science experts

Pew Research Center (ATP)

- Pew conducts nationally representative surveys on a wide range of topics, including politics, social issues, and technology.
- Includes a panel of nationally-representative respondents, used to track trends in public opinion over time

Federal Reserve Economic Data (FRED)

- Federal Reserve is one of the most highly regarded sources for economic data internal to the United States
- Includes data on inflation, GDP, employment levels, at the national and state level

Harvard Dataverse

- Many journals require authors to deposit their data and code in repositories like Harvard Dataverse prior to publication, making it a valuable resource for researchers.
- All data is available for free, and the platform provides tools for searching, accessing, and citing datasets.

Scraping Data

APIs

- Some social media platforms and websites provide APIs (Application Programming Interfaces)
 - Allow developers to access and retrieve data in a structured format.
 - Have limitations on the amount of data you can access and may require authentication.

HTML (HyperText Markup Language)

- Many websites do not provide APIs, but those that are built with HTML can be scraped fairly easily using tools like BeautifulSoup (Python) or rvest (R).
- What is HTML?
 - Consists of a series of elements and tags that define the structure and content of a webpage.

JavaScript

- Many modern websites use JavaScript to dynamically load content, which can make scraping more challenging.
- Tools like Selenium (Python) or RSelenium (R) can be used to automate a web browser and interact with JavaScript elements to extract data.